

CS1231S Condensed Notes · Revision Addendum

This addendum compiles all content added or refined after the initial commit (062d28e), including lecture definitions, equivalence relation notes, well-defined operations on quotient sets, selected tutorial proofs (Part VI), updated index terms, and the complete Appendix A and lecture theorem reference (Part VIII). All material is integrated into the current condensed notes, whose §32 is now a topic locator. This addendum remains a companion for an initial printed copy. Tutorial solutions are adapted from the supplied answers, with explicit domains and justifications.

Part I · Logic & Arguments (§1, §3, §4)

DEFINITION Negation (2.1.2) · Conjunction (2.1.3) · Disjunction (2.1.4) · §1

For statement variables p and q : the **negation** of p is “not p ” or “it is not the case that p ”, denoted $\sim p$; the **conjunction** of p and q is “ p and q ”, denoted $p \wedge q$; the **disjunction** of p and q is “ p or q ”, denoted $p \vee q$ (inclusive: true when either or both are true).

METHOD Clarification on §1 & §3 Wording

- **Tautological / Contradictory statements (§1)**: A statement whose statement form is a tautology (contradiction) is called a **tautological (contradictory)** statement.
- **Conditional terminology (§3)**: In $p \rightarrow q$ (“if p then q ”, “ p implies q ”), p is called the **hypothesis** (or **antecedent**) and q is called the **conclusion** (or **consequent**).

DEFINITION Argument, validity (2.3.1) · Critical row · Syllogism · Sound, unsound (2.3.2) · §4

An **argument** (**argument form**) is a sequence of statements (statement forms); all but the last are **premises** (assumptions, hypotheses), the last (after $\therefore$) is the **conclusion**.

An argument form is **valid** iff, whatever statements are substituted for its variables, if the premises are all true then the conclusion is true.

A **critical row** is a truth-table row in which all premises are T: the form is valid iff the conclusion is T in every critical row.

A **syllogism** is an argument form with two premises and a conclusion (e.g. modus ponens).

An argument is **sound** iff it is valid **and** all its premises are true; otherwise it is **unsound**.

Part II · Predicates, Quantifiers & Proofs (§5, §8, §10)

DEFINITION Domain of discourse & Truth set (§5 update)

The **domain** of a variable is also termed the **domain of discourse**, **universe of discourse**, **universal set**, or **universe**. The **truth set** of $P(x)$, $x \in D$, is the set of all elements of D that make $P(x)$ true: $\{x \in D : P(x)\}$, also denoted $\{x \in D \mid P(x)\}$.

DEFINITION Kinds of mathematical statements (Lecture 1) · §5

- A **universal statement** says a property holds for **all** elements of a set ($\forall$; “all”, “every”, “any”).
- A **conditional statement** says that if one thing is true then another must be ($\rightarrow$; “if ... then”).
- An **existential statement** says there is **at least one** thing with the property ($\exists$; “there exists”, “some”).
- A **universal conditional statement** is both universal and conditional: $\forall x (P(x) \rightarrow Q(x))$ (e.g. “for all animals a , if a is a dog then a is a mammal”).
- A **universal existential statement** has the form $\forall x \exists y \dots$ (e.g. “every real number has an additive inverse”).
- An **existential universal statement** has the form $\exists x \forall y \dots$ (e.g. “there is a positive integer that is $\leq$ every positive integer”).

DEFINITION Valid argument form with quantifiers (3.4.1) · §8

An argument form is **valid** iff, no matter what particular predicates are substituted for the predicate symbols in its premises, if the resulting premise statements are all true then the conclusion is also true. An argument is **valid** iff its form is valid.

DEFINITION Absolute value (Lecture 1) · Colorful (Lecture 1, CS1231S only) · §10

For $x \in \mathbb{R}$, the **absolute value** of x is $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.

An integer n is **colorful** iff there exists some integer k such that $n = 3k$. (Used in lecture: $-1353 = 3 \cdot (-451)$ and $0 = 3 \cdot 0$ are colorful; 7 is not.)

METHOD Axioms & Primes (§10 updates)

- **Axiom / postulate:** Assumed true without proof (Peano axioms, Euclid's postulates).
- **Equivalent prime definition:** $n \in \mathbb{Z}$ with $n > 1$ is prime iff $\forall r, s \in \mathbb{Z} (r > 1 \wedge s > 1 \rightarrow rs \neq n)$.

Part III · Sets & Operations (§13, §14, §16)**DEFINITION Set-roster notation & Cardinality (§13 update)**

- **Set-roster:** list all elements between braces: $\{1, 2, 3\}$, $\{1, 2, \dots, 100\}$, $\{1, 2, 3, \dots\}$ (the ellipsis “...” reads “and so forth”).
- **Set elements:** $x \in S$ (x is a **member** or **element**); $x \notin S$ means it is not.
- **Cardinality** $|S|$: the size of S , the number of elements of S .

DEFINITION Interval notation (Lecture 5) · §13

For real numbers $a \leq b$:

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}, \quad [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\},$$

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}, \quad [a, b) = \{x \in \mathbb{R} : a \leq x < b\}.$$

Unbounded intervals:

$$(a, \infty) = \{x \in \mathbb{R} : x > a\}, \quad [a, \infty) = \{x \in \mathbb{R} : x \geq a\},$$

$$(-\infty, b) = \{x \in \mathbb{R} : x < b\}, \quad (-\infty, b] = \{x \in \mathbb{R} : x \leq b\}.$$

Note: The interval (a, b) and the ordered pair (a, b) look identical; context distinguishes them.

DEFINITION Superset · Singleton (§14 update)

- If $A \subseteq B$, we write $B \supseteq A$: “ B contains / includes A ”, “ B is a **superset** of A ”.
- $A \subsetneq B$ denotes **proper** (strict) inclusion.
- The **empty set** $\emptyset = \{\}$ has no elements; a set with exactly one element is a **singleton**.

DEFINITION Universal set · Relative complement (§16 update)

- A **universal set** (universe of discourse) U contains every object under discussion (e.g. $\mathbb{R}$ when all sets are sets of reals).
- $B \setminus A = \{x \in U : x \in B \wedge x \notin A\}$ is also called the **relative complement** of A in B .

Part IV · Relations, Digraphs & Equivalence (§19, §23)

DEFINITION Relation notation · Directed graph · n -ary relations (§19 update)

- xRy means “ x is R -related to y ”; $x \not R y$ means $(x, y) \notin R$.
- A **(binary) relation on A** is a subset of $A \times A = A^2$; an n -ary relation on A is a subset of A^n .
- The **directed graph** (digraph) of R on A draws each element of A once as a vertex, with a directed arrow $x \rightarrow y$ whenever xRy and a self-loop at x when xRx .
- An **n -ary relation** on $A_1 \times \cdots \times A_n$ is a subset of the Cartesian product (binary, ternary, quaternary for $n = 2, 3, 4$).

DEFINITION Procedural class definition · Congruence modulo n (Lecture 6.3.4) · §23

- Procedural equivalence class: $\forall x \in A (x \in [a]_{\sim} \leftrightarrow a \sim x)$.
- Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$. a is **congruent to b modulo n** , written $a \equiv b \pmod{n}$, iff $a - b = nk$ for some $k \in \mathbb{Z}$ (i.e. $n \mid (a - b)$).
- Classes: $[x] = \{x + nk : k \in \mathbb{Z}\}$, and $\mathbb{Z}/\sim_n = \{[0], [1], \dots, [n-1]\}$.

DEFINITION Representative · Well-defined operation on $A/\sim$ · §23

Any $b \in [a]$ is a **representative** of $[a]$, since then $[b] = [a]$: mod 5, $[2] = [7] = [12]$.

Let $\sim$ be an equivalence relation on A and $*$ a binary operation on A (so $a * b \in A$ for all $a, b \in A$).

The induced rule $[a] * [b] = [a * b]$ is **well defined** iff the answer depends only on the classes, not on the representatives chosen:

$$a \sim a' \wedge b \sim b' \Rightarrow a * b \sim a' * b'.$$

PROOF $+$ and $\cdot$ are well defined on the classes of congruence mod n · §23

1. Fix $n \in \mathbb{Z}^+$ and let $a, a', b, b' \in \mathbb{Z}$ with $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$: $a - a' = nk$ and $b - b' = nm$ for some $k, m \in \mathbb{Z}$ (definition of congruence).
2. (Addition) $(a + b) - (a' + b') = n(k + m)$ with $k + m \in \mathbb{Z}$ (closure), so $a + b \equiv a' + b' \pmod{n}$.
3. (Multiplication) $ab - a'b' = (a - a')b + a'(b - b') = n(kb + a'm)$ with $kb + a'm \in \mathbb{Z}$ (closure), so $ab \equiv a'b' \pmod{n}$. ■

E.g. mod 5: $[2] \cdot [4] = [8] = [3]$ and $[7] \cdot [9] = [63] = [3]$.

TRAP Check reflexive, symmetric, transitive first · then check well-definedness · §23

- $a \sim b \leftrightarrow ab \geq 0$ is **not** transitive on $\mathbb{R}$ ($1 \sim 0$ and $0 \sim -1$, but $1 \not\sim -1$), so it does not give a partition into equivalence classes. On $\mathbb{R} \setminus \{0\}$ it means “same sign”: $(\mathbb{R} \setminus \{0\})/\sim = \{[1], [-1]\} = \{(0, \infty), (-\infty, 0)\}$.
- $a \sim b \leftrightarrow a - b = 2k$ for some $k \in \mathbb{Z}$, on $\mathbb{R}$: $[a] = \{a + 2k : k \in \mathbb{Z}\}$ and $\mathbb{R}/\sim = \{[r] : 0 \leq r < 2\}$ (infinitely many classes). $+$ is well defined, but $\cdot$ is **not**: $[0.5] = [2.5]$, yet $[0.5] \cdot [0.5] = [0.25] \neq [1.25] = [2.5] \cdot [0.5]$, since $1.25 - 0.25 = 1$ is not of the form $2k$.

Part V · Partial Orders & Well-Ordered Sets (§24)

DEFINITION Maximal and minimal alternate characterizations (§24 update)

- c **maximal** $\Leftrightarrow \forall x \in A (c \preceq x \rightarrow c = x)$. Equivalently, every $x \in A$ has $x \preceq c$ or is not comparable with c .
- c **minimal** $\Leftrightarrow \forall x \in A (x \preceq c \rightarrow c = x)$. Equivalently, every $x \in A$ has $c \preceq x$ or is not comparable with c .

DEFINITION Well-ordered set (Lecture 6.4.7) · §24

Let $\preccurlyeq$ be a total order on a set A . A is **well-ordered** iff every non-empty subset of A contains a smallest element. Symbolically,

$$\forall S \in \mathcal{P}(A) (S \neq \emptyset \rightarrow \exists x \in S \forall y \in S (x \preccurlyeq y)).$$

$(\mathbb{N}, \leq)$ is well-ordered. $(\mathbb{Z}, \leq)$ is not: $\mathbb{Z}$ itself (or $\mathbb{Z}^-$) is a non-empty subset with no smallest element.

Theorem: Every total order on a finite set is well-ordered (a non-empty subset has a minimal element, which in a total order is smallest).

Part VI · Tutorial Worked Examples (Tutorials 1–5)

Model solutions adapted from Tutorials 1–5, in the numbered-line, justified style of the main notes. Each illustrates a standard proof pattern. Results already covered by the initial notes ($A \subseteq B \leftrightarrow A \cup B = B$, associativity of $\circ$, $\mathbb{Q}$ closed under $+$, R symmetric $\leftrightarrow R = R^{-1}$, AY23 Q12) are omitted here. The tutorial locator in §29 of the current condensed notes gives the integrated locations.

Proof techniques (Tutorials 1–2)**WORKED EXAMPLE T1 Q11 · Biconditional: n^2 is odd $\leftrightarrow n$ is odd**

For $n \in \mathbb{Z}$, Smart writes: “**Proof (by contradiction)**. Suppose n is even, so $n = 2k$, ... so n^2 is even. So if n is even then n^2 is even, i.e. if n^2 is odd then n is odd. Therefore n^2 is odd iff n is odd.” Comment, then give a correct proof.

Let $n \in \mathbb{Z}$. **Faults:** (1) only one direction (n^2 odd $\rightarrow n$ odd) is proved; (2) the method is **contraposition**, not contradiction; (3) justifications are missing (definition of even, closure).

1. ($\Rightarrow$) Prove the contrapositive of “if n^2 is odd, then n is odd”.
 - 1.1. Suppose n is even. Then $\exists k \in \mathbb{Z}$ s.t. $n = 2k$ (definition of even).
 - 1.2. $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$ (basic algebra), with $2k^2 \in \mathbb{Z}$ (closure of integers under $\times$).
 - 1.3. So n^2 is even (definition of even). This proves: if n^2 is odd, then n is odd (contraposition and the even/odd dichotomy).
2. ($\Leftarrow$) Suppose n is odd: $n = 2k + 1$ for some $k \in \mathbb{Z}$ (definition of odd). Then $n^2 = 2(2k^2 + 2k) + 1$, with $2k^2 + 2k \in \mathbb{Z}$ (basic algebra, closure), so n^2 is odd (definition of odd).
3. Therefore n^2 is odd iff n is odd (lines 1 and 2). ■

WORKED EXAMPLE T2 Q8 · Cases inside a conditional

Prove $\forall x \in \mathbb{R} ((x^2 > x) \rightarrow ((x < 0) \vee (x > 1)))$. Must the case $x^2 \leq x$ be examined?

No: $p \rightarrow q$ is true whenever p is false, so only $x^2 > x$ needs examining.

1. Let r be an arbitrarily chosen real number.
2. Suppose $r^2 > r$.
 - 2.1. Then $r^2 - r > 0$, i.e. $r(r - 1) > 0$ (basic algebra).
 - 2.2. So r and $r - 1$ are both positive or both negative (Appendix A, T25).
 - 2.3. **Case 1:** both positive. Then $r - 1 > 0$, so $r > 1$ (basic algebra).
 - 2.4. **Case 2:** both negative. Then $r < 0$ (case assumption).
 - 2.5. Hence $(r > 1) \vee (r < 0)$; so $(r^2 > r) \rightarrow ((r > 1) \vee (r < 0))$ (division into cases).
3. Therefore $\forall x \in \mathbb{R} ((x^2 > x) \rightarrow ((x < 0) \vee (x > 1)))$ (universal generalization, as r was arbitrary). ■

Do not use diagrams (e.g. graphs) as proofs unless instructed.

PAST PAPER T2 Q11 · Past midterm · A factor is at most $\sqrt{n}$

For $n, a, b \in \mathbb{Z}^+$, prove $n = ab \rightarrow (a \leq \sqrt{n} \vee b \leq \sqrt{n})$. **By contraposition** ($p \rightarrow q \equiv \sim q \rightarrow \sim p$):

1. Contrapositive: if $a > \sqrt{n}$ and $b > \sqrt{n}$, then n is not the product ab (De Morgan’s law).
2. Suppose $a > \sqrt{n}$ and $b > \sqrt{n}$. Then $ab > \sqrt{n} \cdot \sqrt{n} = n$ (Appendix A, T27).
3. Since $ab \neq n$, the contrapositive is true; therefore the original statement is true. ■

By contradiction ($\sim(p \rightarrow q) \equiv p \wedge \sim q$):

1. Suppose not: $n = ab$ and $a > \sqrt{n}$ and $b > \sqrt{n}$ (De Morgan’s law).
2. Then $ab > \sqrt{n} \cdot \sqrt{n} = n$ (Appendix A, T27), contradicting $n = ab$ in line 1. ■

T27: if $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$.

TRAP T2 Q4 · Closure claims

$\mathbb{Z}$ is **not** closed under division ($3/2 = 1.5 \notin \mathbb{Z}$). Division is not a binary operation on **all** of $\mathbb{Q} \times \mathbb{Q}$, since $r/0$ is undefined; division of a rational by a **nonzero** rational does give a rational. In proving $\mathbb{Q}$ closed under $+$, the line “ $bd \neq 0$ (zero product property)” is required, and is commonly omitted.

Sets (Tutorial 3)**WORKED EXAMPLE T3 Q4 · Set equality by double inclusion**

$A = \{2n + 1 : n \in \mathbb{Z}\}$, $B = \{2n - 5 : n \in \mathbb{Z}\}$. Is $A = B$? Prove it.

1. ($\subseteq$) Let $a \in A$.
 - 1.1. Use the definition of A to find an integer n such that $a = 2n + 1$.
 - 1.2. Then $a = 2n + 1 = 2(n + 3) - 5$, and $n + 3 \in \mathbb{Z}$ (closure of integers under $+$).
 - 1.3. Therefore $a \in B$ (definition of B).
2. ($\supseteq$) Let $b \in B$; find $n \in \mathbb{Z}$ with $b = 2n - 5$. Then $b = 2(n - 3) + 1$ with $n - 3 \in \mathbb{Z}$ (closure under $-$), so $b \in A$ (definition of A).
3. Therefore $A = B$ (definition of set equality). ■

The change of variables $k = n + 3$ is valid only after showing it ranges over **all** integers: for each $k \in \mathbb{Z}$, take $n = k - 3 \in \mathbb{Z}$. Closure alone does not justify that step; double inclusion makes both directions explicit.

WORKED EXAMPLE T3 Q5 · Element method as a set-builder chain

Prove that for all sets A, B, C : $A \cap (B \setminus C) = (A \cap B) \setminus C$.

$$\begin{aligned}
 A \cap (B \setminus C) &= \{x : x \in A \wedge x \in (B \setminus C)\} \quad (\text{definition of } \cap) \\
 &= \{x : x \in A \wedge (x \in B \wedge x \notin C)\} \quad (\text{definition of } \setminus) \\
 &= \{x : (x \in A \wedge x \in B) \wedge x \notin C\} \quad (\text{associativity of } \wedge) \\
 &= \{x : x \in A \cap B \wedge x \notin C\} \quad (\text{definition of } \cap) \\
 &= (A \cap B) \setminus C \quad (\text{definition of } \setminus). \quad \blacksquare
 \end{aligned}$$

WORKED EXAMPLE T3 Q6 · Symmetric difference by set identities only

(a) $A = \{1, 4, 9, 16\}$, $B = \{2, 4, 6, \dots, 16\}$: find $A \oplus B$. (b) Using Theorem 6.2.2, prove $A \oplus B = (A \cup B) \setminus (A \cap B)$.

(a) $A \setminus B = \{1, 9\}$, $B \setminus A = \{2, 6, 8, 10, 12, 14\}$, so $A \oplus B = \{1, 2, 6, 8, 9, 10, 12, 14\}$.

$$\begin{aligned}
 (b) \quad A \oplus B &= (A \setminus B) \cup (B \setminus A) \quad (\text{definition of } \oplus) \\
 1. &= (A \cap \overline{B}) \cup (B \cap \overline{A}) \quad (\text{set difference law}) \\
 2. &= ((A \cap \overline{B}) \cup B) \cap ((A \cap \overline{B}) \cup \overline{A}) \quad (\text{distributive law}) \\
 3. &= ((A \cup B) \cap (\overline{B} \cup B)) \cap ((A \cup \overline{A}) \cap (\overline{B} \cup \overline{A})) \quad (\text{distributive, commutative laws}) \\
 4. &= ((A \cup B) \cap U) \cap (U \cap (\overline{A} \cup \overline{B})) \quad (\text{commutative, complement laws}) \\
 5. &= (A \cup B) \cap (\overline{A} \cup \overline{B}) \quad (\text{commutative, identity laws}) \\
 6. &= (A \cup B) \cap \overline{A \cap B} \quad (\text{De Morgan's law}) \\
 7. &= (A \cup B) \setminus (A \cap B) \quad (\text{set difference law}). \quad \blacksquare
 \end{aligned}$$

TRAP T3 Q3(c) · $A \times \mathcal{P}(B) \neq \mathcal{P}(A \times B)$

For every A, B , $\emptyset \in \mathcal{P}(A \times B)$ ($\emptyset \subseteq A \times B$), but $\emptyset \notin A \times \mathcal{P}(B)$ (every member of a Cartesian product is an ordered pair; $\emptyset$ is not one). This named witness proves the sets differ. A cardinality “proof” ($n \cdot 2^k \neq 2^{nk}$ for $|A| = n$, $|B| = k$) is wrong: equality holds when $n = 1$, or $n = 2$ and $k = 1$.

Relations and orders (Tutorials 4–5)**WORKED EXAMPLE T4 Q10 · Equivalence relation using Appendix A**

On $\mathbb{Z} \setminus \{0\}$ let $a \sim b \leftrightarrow ab > 0$. (a) Prove $\sim$ is an equivalence relation. (b) Determine all distinct equivalence classes.

1. (Reflexivity) Let $a \in \mathbb{Z} \setminus \{0\}$. Since $a \neq 0$, $a^2 > 0$ (T21). Thus $a \sim a$ (definition of $\sim$).
2. (Symmetry) Let $a, b \in \mathbb{Z} \setminus \{0\}$ with $a \sim b$. Then $ab > 0$ (definition of $\sim$), so $ba > 0$ (commutative law of $\times$), so $b \sim a$.

3. (Transitivity) Let $a, b, c \in \mathbb{Z} \setminus \{0\}$ with $a \sim b$ and $b \sim c$. Then $ab > 0$ and $bc > 0$ (definition of $\sim$).

3.1. Multiplying, $ab^2c > 0$ (Ord1), so $(ac)b^2 > 0$ (associative, commutative laws of $\times$).

3.2. So ac and b^2 are both positive or both negative (T25).

3.3. Since $b^2 > 0$ (T21, as $b \neq 0$), $ac > 0$. Thus $a \sim c$.

4. Therefore $\sim$ is an equivalence relation. ■

(b) T25 rules out pairs of opposite signs; the product of two positive or two negative integers is positive. The classes are $[1] = \mathbb{Z}^+$ and $[-1] = \mathbb{Z}^-$, so $(\mathbb{Z} \setminus \{0\})/\sim = \{\mathbb{Z}^+, \mathbb{Z}^-\}$.

T21: $a \neq 0 \Rightarrow a^2 > 0$. Ord1: $a, b > 0 \Rightarrow a + b > 0 \wedge ab > 0$. T25: $ab > 0 \Rightarrow a, b$ both positive or both negative.

WORKED EXAMPLE T4 Q9 · Inverse and composition by the element method

$S = \{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \text{ is even}\}$. Determine (a) S^{-1} , (b) $S \circ S$, (c) $S \circ S^{-1}$.

(a) $S^{-1} = \{(x, y) \in \mathbb{Z}^2 : (y, x) \in S\}$ (definition of inverse)

$= \{(x, y) \in \mathbb{Z}^2 : y^2 + x^2 \text{ is even}\}$ (definition of S)

$= \{(x, y) \in \mathbb{Z}^2 : x^2 + y^2 \text{ is even}\}$ (commutativity of $+$)

$= S$.

(b) $S \circ S = S$:

1. ($\subseteq$) Let $(x, z) \in S \circ S$. Then $(x, y) \in S$ and $(y, z) \in S$ for some $y \in \mathbb{Z}$ (definition of composition).

1.1. So $x^2 + y^2$ and $y^2 + z^2$ are even (definition of S), hence $x^2 + 2y^2 + z^2$ is even (sum of two even integers).

1.2. As $2y^2$ is even (definition of even, closure), $x^2 + z^2$ is even (difference of two even integers), so $(x, z) \in S$ (definition of S).

2. ($\supseteq$) Let $(x, z) \in S$. Note $(x, x) \in S$, as $x^2 + x^2$ is even. Since $(x, x) \in S$ and $(x, z) \in S$, $(x, z) \in S \circ S$ (definition of composition).

3. Therefore $S \circ S = S$. ■

(c) $S \circ S^{-1} = S \circ S = S$ by (a) and (b).

Alternative for ($\supseteq$): by cases. If x^2 is odd, so is z^2 , and $(x, 1), (1, z) \in S$. If x^2 is even, so is z^2 , and $(x, 0), (0, z) \in S$.

PAST PAPER T4 Q8 (AY20/21 exam) · T4 Q6 (AY23/24 exam)

Q8. On $\mathbb{Z}^+ \times \mathbb{Z}^+$, $(a, b) \sim (c, d) \leftrightarrow ab = cd$. Write the set-builder line first: $[(1, 1)]$

$= \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : (1, 1) \sim (x, y)\}$ (definition of class)

$= \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : xy = 1\}$ (definition of $\sim$)

$= \{(1, 1)\}$. Likewise $[(4, 3)] = \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : xy = 12\} = \{(1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1)\}$.

Q6. $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$ on $\{1, 2, 3\}$; find $R \circ R \circ \dots \circ R$ (7 copies). R is an equivalence relation, so $R \circ R = R$ (AY23 Q12, §23). By associativity, $R^7 = ((R \circ R) \circ (R \circ R)) \circ ((R \circ R) \circ R) = (R \circ R) \circ (R \circ R) = R \circ R = R$.

WORKED EXAMPLE T5 Q6 · Reflexive closure (the closure-proof template)

Let R be a relation on a non-empty set A , and define S on A by $xSy \leftrightarrow ((x = y) \vee (xRy))$. Show: (a) S is reflexive; (b) $R \subseteq S$; (c) if S' is a reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$. What is S called?

(a) Let $x \in A$. $x = x$, so xSx (definition of S). Therefore S is reflexive.

(b) Suppose $(x, y) \in R$. Then xSy (definition of S), so $(x, y) \in S$. Therefore $R \subseteq S$ (definition of $\subseteq$).

(c) Let $(x, y) \in S$: $x = y$ or xRy (definition of S).

1. **Case 1 ($x = y$):** $xS'y$ (S' is reflexive), so $(x, y) \in S'$.

2. **Case 2 (xRy):** $(x, y) \in R \subseteq S'$, so $(x, y) \in S'$.

In all cases $(x, y) \in S'$. Therefore $S \subseteq S'$. ■

S is the **reflexive closure** of R : the smallest reflexive relation on A containing R . Parts (a)–(c) mirror the three conditions defining a closure.

WORKED EXAMPLE T5 Q7 · Asymmetric vs antisymmetric

On $A = \{a, b, c\}$ find a relation that is (a) asymmetric and antisymmetric; (b) antisymmetric but not asymmetric; (c) asymmetric but not antisymmetric; (d) neither.

(a) $\{(a, b), (a, c)\}$. (b) $\{(a, a), (b, c)\}$. (c) **No solution:** every asymmetric relation is antisymmetric (below). (d) $\{(a, b), (b, a), (a, c)\}$.

1. Let R be an asymmetric relation on A : $\forall x, y \in A (xRy \rightarrow yRx)$ (definition of asymmetry).
 - 1.1. $\equiv \forall x, y \in A (xRy \vee yRx)$ (implication law).
 - 1.2. $\Rightarrow \forall x, y \in A ((xRy \vee yRx) \vee x = y)$ (generalization).
 - 1.3. $\equiv \forall x, y \in A (\sim(xRy \wedge yRx) \vee x = y)$ (De Morgan's law).
 - 1.4. $\equiv \forall x, y \in A (xRy \wedge yRx \rightarrow x = y)$ (implication law).
2. The final formula is the definition of antisymmetry, so R is antisymmetric: asymmetry makes it vacuously true. ■

TRAP T5 Q1 · Disprove with one named counterexample

On strings over $\{s, u\}$, $aRb \leftrightarrow \text{len}(a) \leq \text{len}(b)$ is **not** a partial order: sRu and uRs but $s \neq u$, so R is not antisymmetric. One counterexample suffices; there is no need to check reflexivity or transitivity.

Part VII · Definition locator for the original notes

Absolute value: $|x| = x$ if $x \geq 0$, $-x$ if $x < 0$. §10

Argument; valid; sound, unsound: premises then conclusion; valid if true premises force a true conclusion; sound if valid with true premises, else unsound. §4

Colorful: $n = 3k$ for some $k \in \mathbb{Z}$ (lecture only). §10

Conditional: $p \rightarrow q$, false only for T $\rightarrow$ F; hypothesis p , conclusion q . §3

Critical row: truth-table row with all premises true. §4

Directed graph: one vertex per element; arrow $x \rightarrow y$ iff xRy . §19

Disjoint; mutually disjoint: $A \cap B = \emptyset$; $A_i \cap A_j = \emptyset$ for $i \neq j$. §16

Empty set; singleton: $\emptyset$, no elements; exactly one element. §14

Interval notation: (a, b) , $[a, b]$, $(a, b]$, $[a, b)$, $[a, \infty)$, ... as subsets of $\mathbb{R}$. §13

Lowest terms: a/b where 1 is the largest common divisor. §10

n -ary relation: subset of $A_1 \times \dots \times A_n$. §19

Negation, conjunction, disjunction: $\sim p$; $p \wedge q$; $p \vee q$. §1

Ordered pair; n -tuple: equal iff equal componentwise. §15

Partition; component: non-empty subsets; each element in exactly one; its elements. §18

Predicate; domain; truth set: sentence with variables; allowed values; $\{x \in D : P(x)\}$. §5

Rational / irrational: $r = a/b$ with $a, b \in \mathbb{Z}$, $b \neq 0$ / not rational. §10

Reflexive / symmetric closure: $R \cup \{(x, x) : x \in A\}$ / $R \cup R^{-1}$. §22

Relation; relation on A : subset of $A \times B$; subset of $A \times A$. §19

Representative: any member b of $[a]$; then $[b] = [a]$. §23

Set equality: $A \subseteq B$ and $B \subseteq A$. §14

Set-roster notation: elements listed in braces. §13

Subset; superset: $\forall x(x \in A \rightarrow x \in B)$; $B \supseteq A$. §14

Syllogism: two premises and a conclusion. §4

Uniqueness $\exists!$: exactly one element satisfies it. §5

Universal conditional; universal existential; existential universal: $\forall x(P \rightarrow Q)$; $\forall \exists$; $\exists \forall$. §5

Universal set: U , all objects under discussion. §16

Valid argument form: true premises force a true conclusion. §4, §8

Well-defined operation: binary operation on A inducing a rule on classes independent of representatives. §23

Well-ordered: total order in which every non-empty subset has a smallest element. §24

Part VIII · Appendix A and Lecture Theorems (§33–34)

Appendix A · Properties of the Real Numbers

Source: Appendix A, cited in Lecture 1, slide 26; Lecture 3, slide 81; and Lecture 4, slide 4. The complete appendix is reproduced here as a statement reference, including its final least upper bound axiom. The symbols a, b, c, d below denote arbitrary **real numbers**, subject to each stated condition.

Appendix A · Field axioms F1–F6

Addition and multiplication are binary operations on $\mathbb{R}$: if $a, b \in \mathbb{R}$, then $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$.

Appendix A label	Statement
Axiom F1 · Commutative laws	$a + b = b + a$; $ab = ba$.
Axiom F2 · Associative laws	$(a + b) + c = a + (b + c)$; $(ab)c = a(bc)$.
Axiom F3 · Distributive laws	$a(b + c) = ab + ac$; $(b + c)a = ba + ca$.
Axiom F4 · Identity elements	There are distinct $0, 1 \in \mathbb{R}$ such that $0 + a = a + 0 = a$ and $1 \cdot a = a \cdot 1 = a$.
Axiom F5 · Additive inverses	For each a , there is $-a \in \mathbb{R}$ with $a + (-a) = (-a) + a = 0$.
Axiom F6 · Reciprocals	For each $a \neq 0$, there is $a^{-1} = 1/a \in \mathbb{R}$ with $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

Appendix A · Theorems T1–T16 (algebra)

Appendix A label	Statement
Theorem T1 · Addition cancellation	If $a + b = a + c$, then $b = c$. The additive identity 0 is unique.
Theorem T2 · Subtraction	For each a, b , exactly one x satisfies $a + x = b$. This x is denoted $b - a$; $0 - a = -a$.
Theorem T3	$b - a = b + (-a)$.
Theorem T4	$-(-a) = a$.
Theorem T5	$a(b - c) = ab - ac$.
Theorem T6	$0 \cdot a = a \cdot 0 = 0$.
Theorem T7 · Multiplication cancellation	If $ab = ac$ and $a \neq 0$, then $b = c$. The multiplicative identity 1 is unique.
Theorem T8 · Division	For $a \neq 0$, exactly one x satisfies $ax = b$. This x is denoted b/a ; $1/a$ is the reciprocal of a .
Theorem T9	If $a \neq 0$, then $b/a = b \cdot a^{-1}$.
Theorem T10	If $a \neq 0$, then $(a^{-1})^{-1} = a$.
Theorem T11 · Zero product	If $ab = 0$, then $a = 0$ or $b = 0$.
Theorem T12 · Negative signs	$(-a)b = a(-b) = -(ab)$; $(-a)(-b) = ab$. If $b \neq 0$, $-(a/b) = (-a)/b = a/(-b)$.
Theorem T13 · Equivalent fractions	If $b \neq 0$ and $c \neq 0$, then $a/b = (ac)/(bc)$.
Theorem T14 · Addition of fractions	If $b \neq 0$ and $d \neq 0$, then $a/b + c/d = (ad + bc)/(bd)$.
Theorem T15 · Multiplication of fractions	If $b \neq 0$ and $d \neq 0$, then $(a/b)(c/d) = (ac)/(bd)$.
Theorem T16 · Division of fractions	If $b \neq 0$, $c \neq 0$ and $d \neq 0$, then $(a/b)/(c/d) = (ad)/(bc)$.

Appendix A · Order axioms Ord1–Ord3

Appendix A label	Statement
Axiom Ord1	If a and b are positive, then $a + b$ and ab are positive.
Axiom Ord2	If $a \neq 0$, exactly one of a and $-a$ is positive.
Axiom Ord3	0 is not positive.

Appendix A · Order definitions: $a < b$ means $b + (-a)$ is positive; $b > a$ means $a < b$. Also, $a \leq b$ means $a < b$ or $a = b$, and $b \geq a$ means $a \leq b$. A number a is **negative** if $a < 0$, and **nonnegative** if $a \geq 0$.

Appendix A · Theorems T17–T27 (order)

Appendix A label	Statement
Theorem T17 · Trichotomy	Exactly one holds: $a < b$, $b < a$, or $a = b$.
Theorem T18 · Transitivity	If $a < b$ and $b < c$, then $a < c$.
Theorem T19	If $a < b$, then $a + c < b + c$.
Theorem T20	If $a < b$ and $c > 0$, then $ac < bc$.
Theorem T21	If $a \neq 0$, then $a^2 > 0$.
Theorem T22	$1 > 0$.
Theorem T23	If $a < b$ and $c < 0$, then $ac > bc$.
Theorem T24	If $a < b$, then $-a > -b$. In particular, if $a < 0$, then $-a > 0$.
Theorem T25	If $ab > 0$, then a and b are both positive or both negative.
Theorem T26	If $a < c$ and $b < d$, then $a + b < c + d$.
Theorem T27	If $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$.

AXIOM Appendix A · Axiom LUB · Least upper bound

Every nonempty subset S of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$. Explicitly, let $B = \{u \in \mathbb{R} : \forall s \in S, s \leq u\}$.

If $S \neq \emptyset$ and $B \neq \emptyset$, then there is $\ell \in B$ such that $\forall u \in B, \ell \leq u$. This ℓ is the **least upper bound** (supremum) of S . This axiom holds for $\mathbb{R}$; the analogous statement for subsets and bounds in $\mathbb{Q}$ fails.

Lecture 4, slide 4 · Additional assumptions: equality is reflexive, symmetric and transitive for all objects; no integer lies strictly between 0 and 1; and $\mathbb{Z}$ is closed under addition, subtraction and multiplication. Lecture 1 additionally permits the even/odd dichotomy (Assumption 1, slide 27) and reduction of every rational to lowest terms (Assumption 2, slide 37).

Lecture theorem reference

Numbering: the primary labels below are the labels in the supplied lecture slides. Lemma and proposition labels are preserved. Results with no number in the slides are identified by lecture and slide.

Theorem 2.1.1 · Logical equivalences

Source: Lecture 2, slides 25–26. Let p, q, r be statement variables, t a tautology, and c a contradiction.

Law in Theorem 2.1.1	Equivalences
1 · Commutative	$p \wedge q \equiv q \wedge p$; $p \vee q \equiv q \vee p$.
2 · Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$; $(p \vee q) \vee r \equiv p \vee (q \vee r)$.
3 · Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$; $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
4 · Identity	$p \wedge t \equiv p$; $p \vee c \equiv p$.
5 · Negation	$p \vee \sim p \equiv t$; $p \wedge \sim p \equiv c$.
6 · Double negative	$\sim(\sim p) \equiv p$.
7 · Idempotent	$p \wedge p \equiv p$; $p \vee p \equiv p$.
8 · Universal bound	$p \vee t \equiv t$; $p \wedge c \equiv c$.
9 · De Morgan	$\sim(p \wedge q) \equiv \sim p \vee \sim q$; $\sim(p \vee q) \equiv \sim p \wedge \sim q$.
10 · Absorption	$p \vee (p \wedge q) \equiv p$; $p \wedge (p \vee q) \equiv p$.
11 · Negation of true/false	$\sim t \equiv c$; $\sim c \equiv t$.

Theorems 3.2.1 and 3.2.2 · Quantifier negation

THEOREM Theorem 3.2.1 · Negation of a universal statement

$$\sim(\forall x \in D, P(x)) \equiv \exists x \in D, \sim P(x).$$

“Not all satisfy P ” means “some do not satisfy P ”. **Source:** Lecture 3, slide 29.

THEOREM Theorem 3.2.2 · Negation of an existential statement

$$\sim(\exists x \in D, P(x)) \equiv \forall x \in D, \sim P(x).$$

“None satisfy P ” means “all fail P ”. **Source:** Lecture 3, slide 30.

Number theory · Slide-numbered results

THEOREM Theorem 4.2.1 · Every integer is rational

For every $n \in \mathbb{Z}$, $n \in \mathbb{Q}$. **Source:** Lecture 4, slide 19.

THEOREM Theorem 4.2.2 · Sum of rational numbers

For every $r, s \in \mathbb{Q}$, $r + s \in \mathbb{Q}$. **Sources:** Lecture 1, slide 25; Lecture 4, slide 20.

COROLLARY Corollary 4.2.3 · Double of a rational number

For every $r \in \mathbb{Q}$, $2r \in \mathbb{Q}$. **Sources:** Lecture 1, slide 25; Lecture 4, slide 21.

THEOREM Theorem 4.3.1 · A positive divisor of a positive integer

For every $a, b \in \mathbb{Z}^+$, if $a \mid b$, then $a \leq b$. **Source:** Lecture 4, slide 24.

THEOREM Theorem 4.3.2 · Divisors of 1

The only integer divisors of 1 are 1 and -1 . **Source:** Lecture 4, slide 25.

THEOREM Theorem 4.3.3 · Transitivity of divisibility

For every $a, b, c \in \mathbb{Z}$, if $a \mid b$ and $b \mid c$, then $a \mid c$. **Sources:** Lecture 4, slide 26; Lecture 6, slides 69–70.

THEOREM Theorem 4.4.1 · Quotient–Remainder Theorem

For every $n \in \mathbb{Z}$ and $d \in \mathbb{Z}^+$, there are unique $q, r \in \mathbb{Z}$ such that $n = dq + r$ and $0 \leq r < d$.

Source: Lecture 5, slide 32.

LEMMA Lemma 4.4.4 · Absolute value bounds

For every $r \in \mathbb{R}$, $-|r| \leq r \leq |r|$. **Source:** Lecture 1, slide 24.

THEOREM Theorem 4.4.6 · Triangle inequality

For every $x, y \in \mathbb{R}$, $|x + y| \leq |x| + |y|$. **Source:** Lecture 1, slide 24.

THEOREM Theorem 4.6.1 · There is no greatest integer

For every $n \in \mathbb{Z}$, there is $m \in \mathbb{Z}$ with $m > n$. **Source:** Lecture 4, slide 29.

PROPOSITION Proposition 4.6.4 · Even square implies even integer

For every $n \in \mathbb{Z}$, if n^2 is even, then n is even. **Sources:** Lecture 1, slide 38; Lecture 4, slides 31–32.

THEOREM Theorem 4.7.1 · Irrationality of the square root of 2

$\sqrt{2} \notin \mathbb{Q}$. **Sources:** Lecture 1, slides 36, 39; Lecture 4, slide 28.

Set theory · Slide-numbered results**THEOREM Theorem 6.2.1 · Some subset relations**

For all sets A, B, C :

- **Inclusion of intersection:** $A \cap B \subseteq A$ and $A \cap B \subseteq B$.
- **Inclusion in union:** $A \subseteq A \cup B$ and $B \subseteq A \cup B$.
- **Transitivity:** if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Source: Lecture 5, slide 40.

Theorem 6.2.2 · Set identities

Source: Lecture 5, slides 42–43. All sets below are subsets of the same universal set U .

Law in Theorem 6.2.2	Identities
1 · Commutative	$A \cup B = B \cup A$; $A \cap B = B \cap A$.
2 · Associative	$(A \cup B) \cup C = A \cup (B \cup C)$; $(A \cap B) \cap C = A \cap (B \cap C)$.
3 · Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$; $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
4 · Identity	$A \cup \emptyset = A$; $A \cap U = A$.
5 · Complement	$A \cup \overline{A} = U$; $A \cap \overline{A} = \emptyset$.
6 · Double complement	$\overline{\overline{A}} = A$.
7 · Idempotent	$A \cup A = A$; $A \cap A = A$.
8 · Universal bound	$A \cup U = U$; $A \cap \emptyset = \emptyset$.
9 · De Morgan	$\overline{A \cup B} = \overline{A} \cap \overline{B}$; $\overline{A \cap B} = \overline{A} \cup \overline{B}$.
10 · Absorption	$A \cup (A \cap B) = A$; $A \cap (A \cup B) = A$.
11 · Complements of U , $\emptyset$	$\overline{\overline{U}} = \emptyset$; $\overline{\emptyset} = U$.
12 · Set difference	$A \setminus B = A \cap \overline{B}$.

THEOREM Theorem 6.2.4 · The empty set

For every set A , $\emptyset \subseteq A$. **Source:** Lecture 5, slide 15.

THEOREM Theorem 6.3.1 · Cardinality of a power set

If A is a **finite** set with $|A| = n$, then $|\mathcal{P}(A)| = 2^n$. **Source:** Lecture 5, slides 34–36 (the named theorem on slide 34 receives this number on slide 36).

Relations · Slide-numbered results**THEOREM Theorem 8.3.1 · Relation induced by a partition**

If $\mathcal{C}$ is a partition of A and xRy means that x and y lie in the same component of $\mathcal{C}$, then R is reflexive, symmetric and transitive, hence an equivalence relation on A . **Source:** Lecture 6, slide 39.

LEMMA Lemma Rel.1 · Equivalence classes

Let $\sim$ be an equivalence relation on A . For every $x, y \in A$, the following are equivalent:

1. $x \sim y$.
2. $[x]_{\sim} = [y]_{\sim}$.
3. $[x]_{\sim} \cap [y]_{\sim} \neq \emptyset$.

Source: Lecture 6, slides 47–49.

THEOREM Theorem 8.3.4 · Theorem Rel.2 · Equivalence classes form a partition

If $\sim$ is an equivalence relation on A , its distinct equivalence classes form a partition of A :

$$A/\sim = \{[x]_{\sim} : x \in A\}.$$

Each class is nonempty, their union is A , and distinct classes are disjoint. **Sources:** Lecture 6, slide 50 (Theorem 8.3.4), slides 57–59 (Theorem Rel.2). These are two slide labels for the same result.

Propositions without theorem numbers in the slides**PROPOSITION Proposition · Composition is associative (Lecture 6, slide 18)**

Let $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$. Then

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

PROPOSITION Proposition · Inverse of composition (Lecture 6, slide 18)

Let $R \subseteq A \times B$ and $S \subseteq B \times C$. Then

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}.$$

PROPOSITION Proposition · Congruence (Lecture 6, slide 54)

For every $n \in \mathbb{Z}^+$, the relation $a \equiv b \pmod{n}$ defined by $n \mid (a - b)$ is an equivalence relation on $\mathbb{Z}$.

PROPOSITION Proposition · A smallest element is minimal (Lecture 6, slide 83)

In any poset $(A, \preceq)$, every smallest element is minimal. Likewise, every largest element is maximal.